HCI Lecture Notes: Case Studies from the Animation & Gaming Industry

Course: Human-Computer Interaction (HCI)
Topic: Case Studies from the Animation & Gaming Industry

1. Introduction

Animation and gaming are among the largest industries where Human-Computer Interaction (HCI) plays a crucial role.
A beautiful game with poor usability often fails, whereas a game with excellent interaction design becomes highly successful.
HCI ensures that users can:

- Learn controls easily
- Navigate interfaces efficiently
- Receive immediate feedback
- Enjoy immersive experiences
- Stay engaged for long periods

Importance of HCI in Games

Good HCI provides:

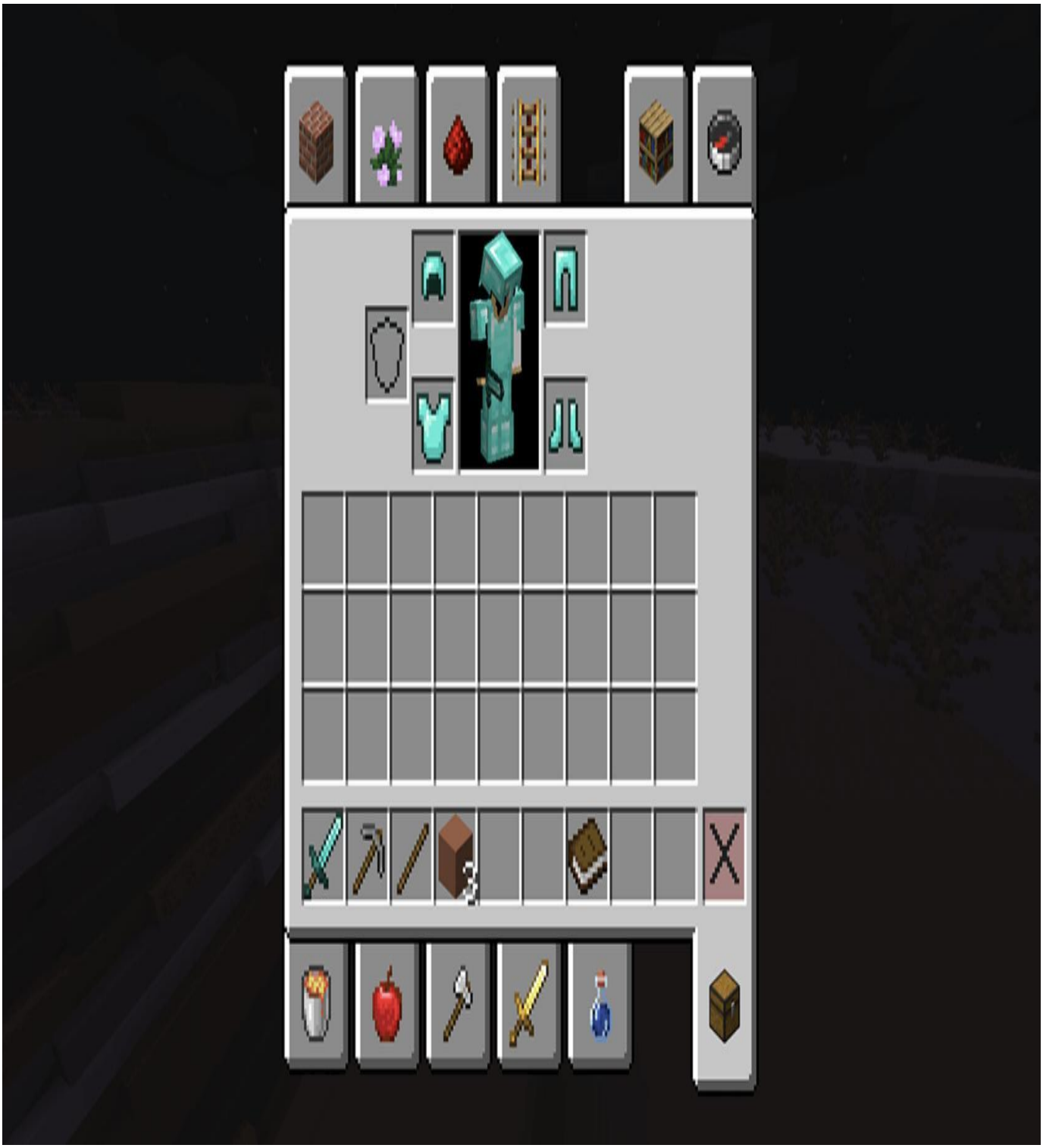
- Easy navigation
- Responsive controls
- Clear feedback
- Accessibility
- High player satisfaction
- Better learning curve
- Reduced user frustration

HCI Components in Games

Component	Example
User Interface (UI)	Health bar, menus
User Experience (UX)	Fun and engagement
Feedback	Sound, vibration, animations
Accessibility	Color-blind mode
Interaction	Keyboard, controller, touch, VR

Case Study 1: Minecraft

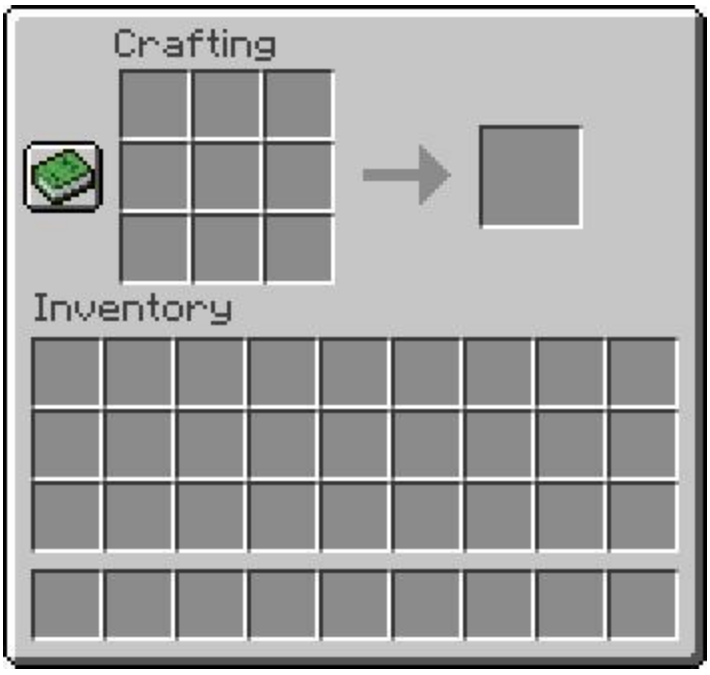





Crafting
Inventory


Status Icons
Armour
Health
Hunger
Experience
Off Hand
Toolbar


Crafting
9
24
41
2
34
43
20
15
19
37



Overview

Minecraft is a sandbox game where players explore, build, and survive.

HCI Features

Simple Interface

Only essential information appears.

- Health
- Hunger
- Inventory

Minimal interface reduces distraction.

Consistent Controls

- Left click → Break blocks
- Right click → Place blocks

Users learn quickly because controls remain consistent.

Immediate Feedback

Actions produce instant feedback.
Example:
Breaking a block results in:

- Animation
- Sound
- Particle effects

Discoverability

Players learn through exploration rather than lengthy tutorials.

HCI Principles Used

- Visibility
- Feedback
- Learnability
- Consistency
- Simplicity

Success Factors

- Easy to learn
- Suitable for all ages
- Encourages creativity

Case Study 2: Fortnite




INVENTORY
60
34
37
51
41
9
x10
Common | World Resource
WOOD
Basic resource found throughout the world.
Backpack 5/5
9
51
41
37
34
35
10
6
29
39
10
9
ESC Back
E Equip
X Drop


Accessibility
Text to Speech
Speak text on the screen, and provide spoken guidance for operating the system. Text to speech works only with some features.
Zoom
Invert Colors
Larger Text
Bold Text
High Contrast
Closed Captions
Button Assignments
Enter
Back


madridbros
Xilien Zero
Bunnichas
NE
E
SE
S
FIELDS
0:57
28
0
XilienZero
5
5
99
0
20
QuiKSor_68 eliminated
with a rifle
30 31
65 100
75 100
Hold


briocheen
123
E
SE
105
120
150
165
artenisa2000
Uncommon | Ranged Weapon
TACTICAL SHOTGUN
Ranged | Shotgun
LV 20
DPS 112.5
Damage 75.0
Fire Rate 1.5
Magazine Size 8
Reload Time 6.0
Traps & Resources
212
215
21
Ammo
40
18
90
6
34
Swap
Equipment
42
3
41
8
50
2 6
100 100
100 100
Z Split
X Drop
ESC Back

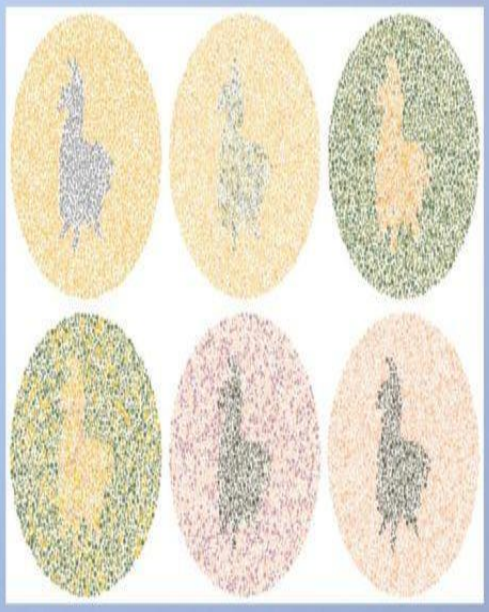
Settings

Video Game Brightness Audio Accessibility Input Controller

EXPERIMENTAL COLOR CORRECTION

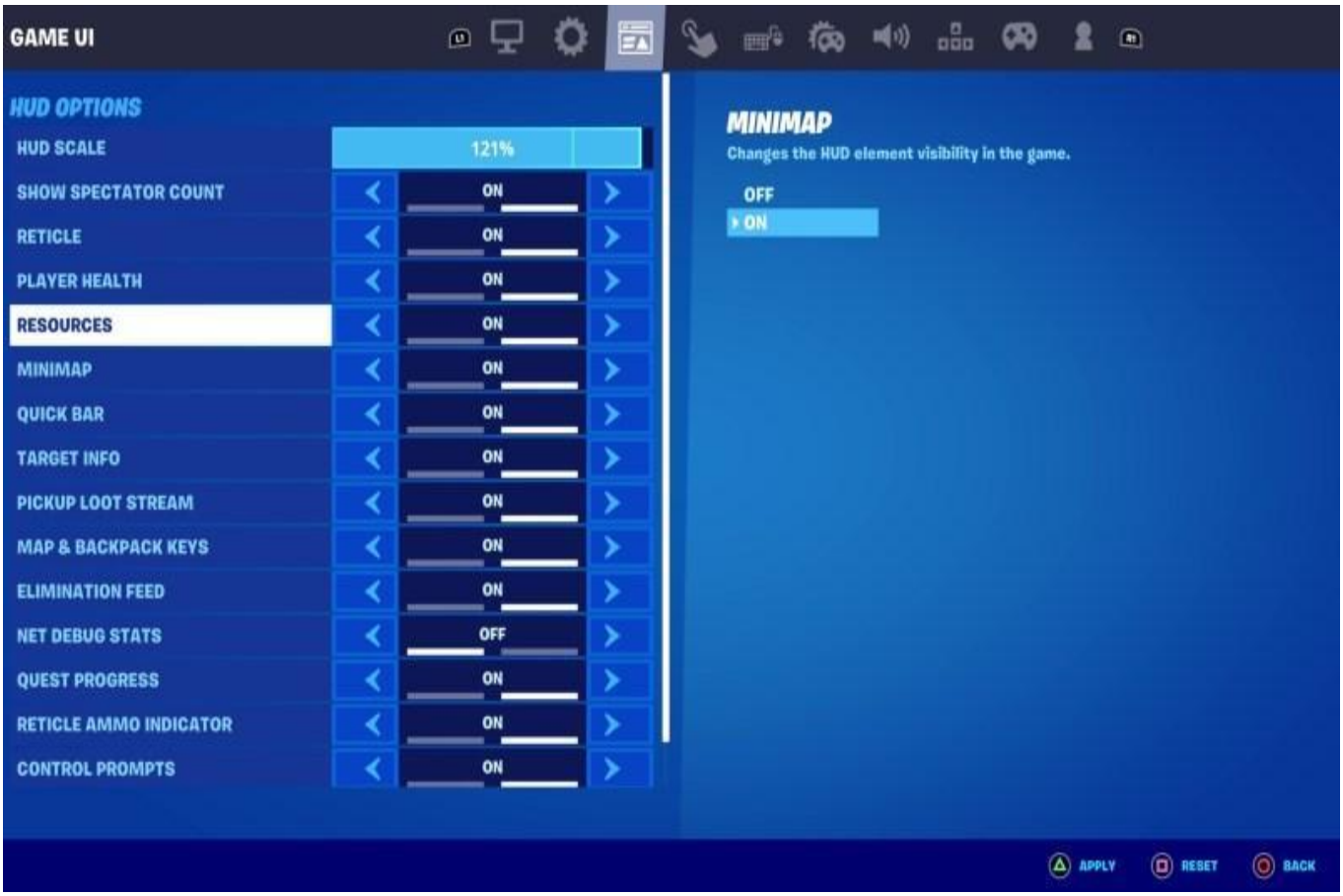
Color Blind Mode ◀ Deuteranope ▶

Color Blind Strength 10



Global Currently unavailable.

A Apply R Restore ESC Back



Overview

Fortnite combines shooting, survival, and building mechanics.

HCI Features

Dynamic HUD

Displays

- Health
- Shield
- Ammo
- Map
- Building materials

without overwhelming the player.

Visual Feedback

- Hit markers
- Damage numbers
- Building animations
- Color indicators

Accessibility

Fortnite includes

- Color-blind mode
- Subtitle customization
- Controller remapping
- Adjustable sensitivity
- Visualized sound effects

Social Interaction
Players communicate using
- Voice chat
- Emotes
- Ping system

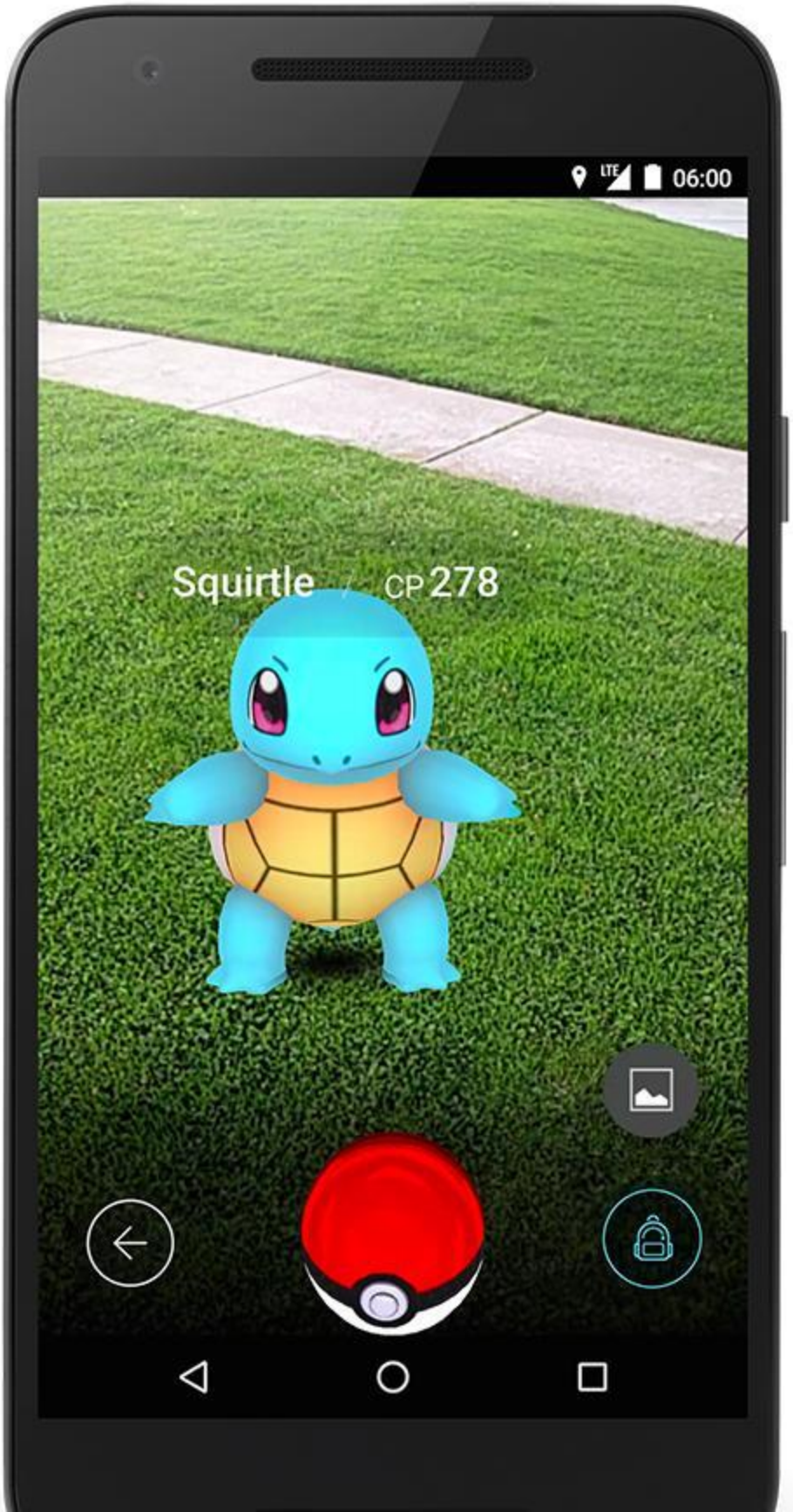
HCI Principles
- Flexibility
- Accessibility
- Immediate feedback
- User control

Case Study 3: Pokémon GO (AR Game)


06:00
Squirtle / CP 268


70%
1:13
5
SteepsM


Pikachu
CP???


06:00
Squirtle
CP 278


11:07
CHECK OUT A POKÉSTOP


SAMSUNG
57% 19:16
AR
50



Overview

Pokémon GO combines augmented reality with GPS.

HCI Features

Natural Interaction

Users

- Walk
- Rotate phone
- Tap screen

instead of using complex controls.

Real-Time Feedback

- Pokémon appears instantly.
- Vibrations confirm actions.
- Animations increase engagement.

Context Awareness

The game uses

- GPS
- Camera
- Time
- Weather

to change gameplay.

HCI Principles

- Natural interaction
- Context-aware computing
- Visibility
- Real-world mapping

Challenges

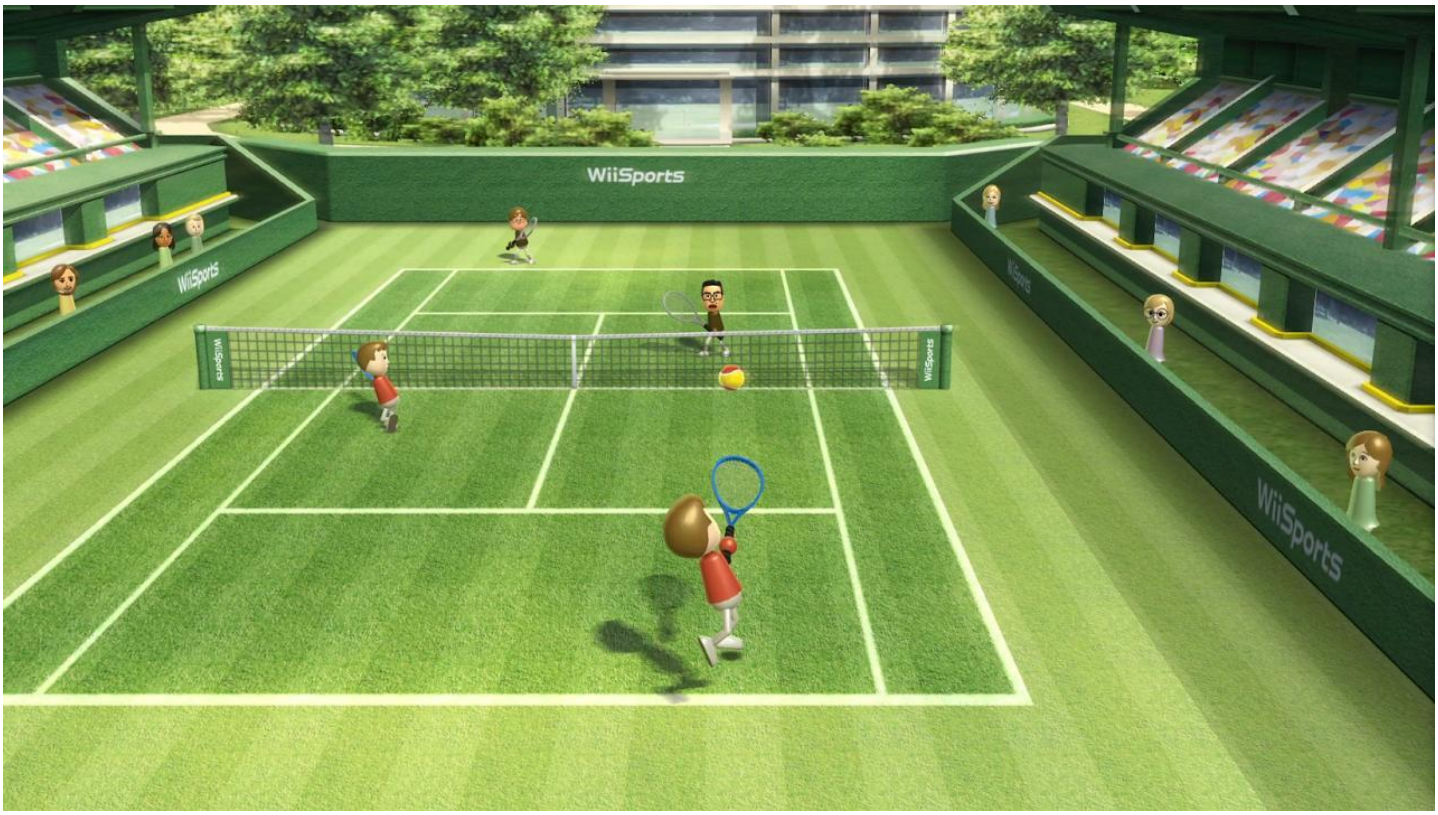
- Battery usage
- Safety while walking
- GPS accuracy

Case Study 4: Wii Sports




A
HOME
1
2
Wii


7 8 9 10
4 5 6
2 3
1
1 2 3
1
9
移動
A
切替


WiiSports
WiiSports
WiiSports
WiiSports
WiiSports


POWER
A
HOME
1
2
Wii


4
2
1
2
3
4
6 - 4 - 7
22
28

Overview
Nintendo introduced motion-based gaming.
Players physically move instead of pressing buttons.

HCI Features
Gesture-Based Interaction
Examples:
- Swing controller → Tennis
- Bowl motion → Bowling

Immediate Feedback
The screen reflects every movement instantly.

Inclusive Design
Suitable for
- Children
- Adults
- Elderly users

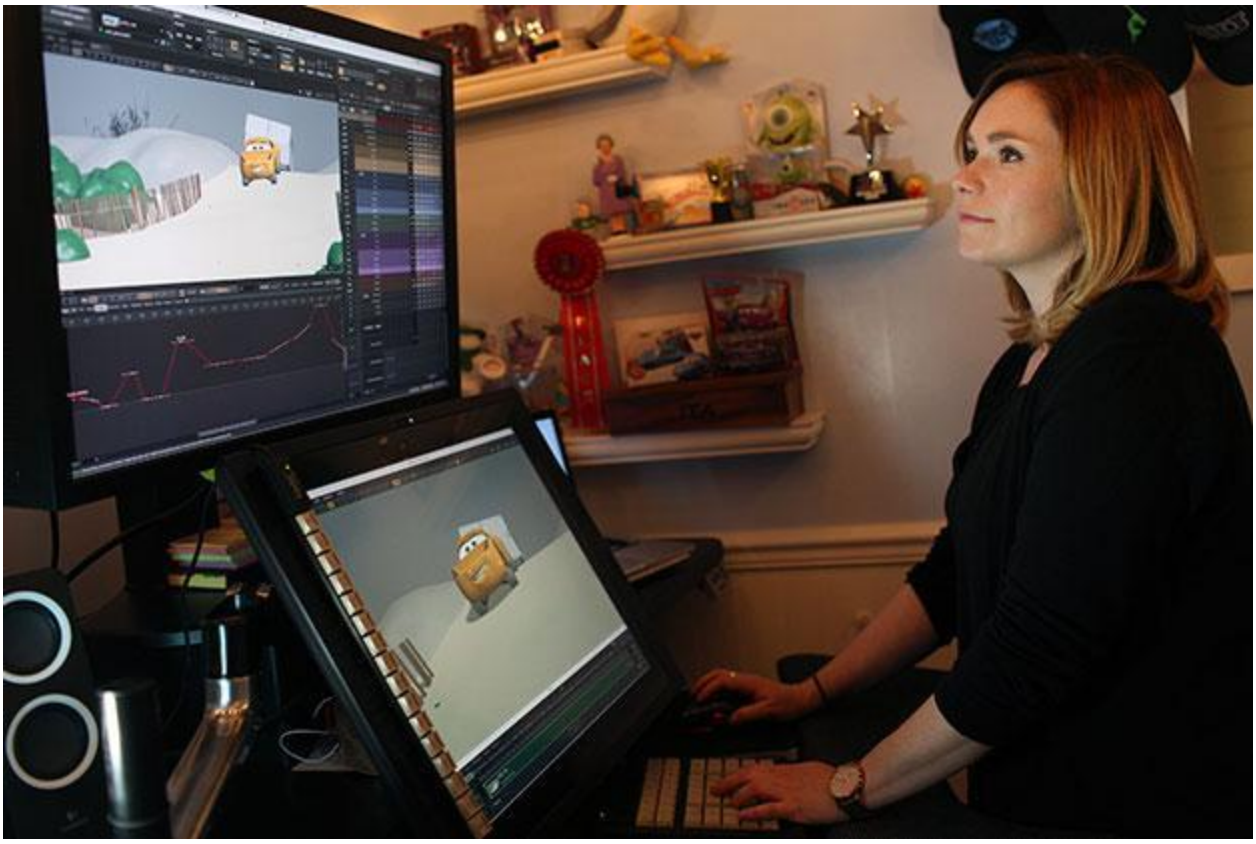
HCI Principles
- Natural mapping
- Direct manipulation
- Learnability
- Accessibility

Success
Millions of first-time gamers learned gaming through Wii Sports.

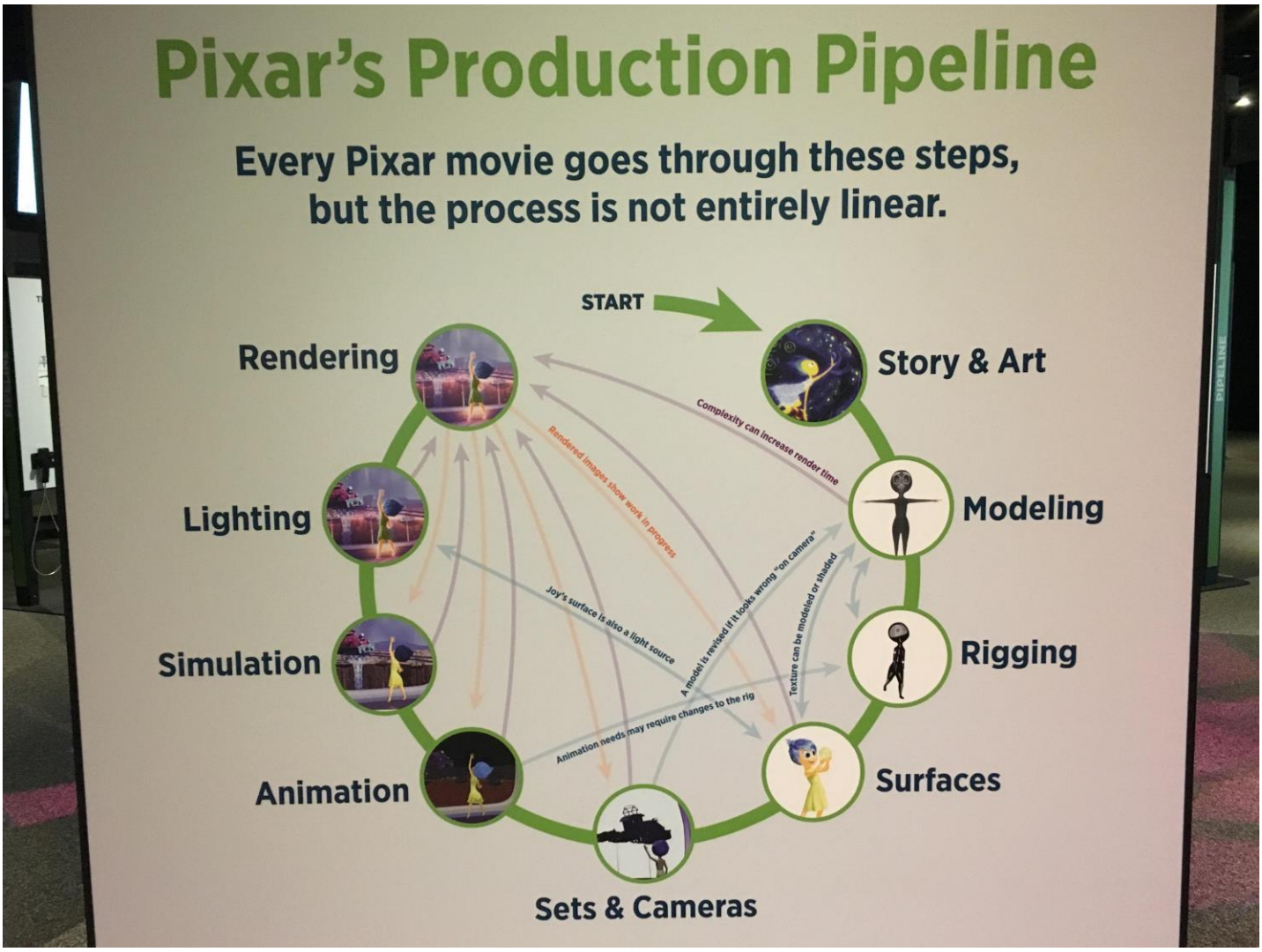
Case Study 5: Pixar Animation Studio

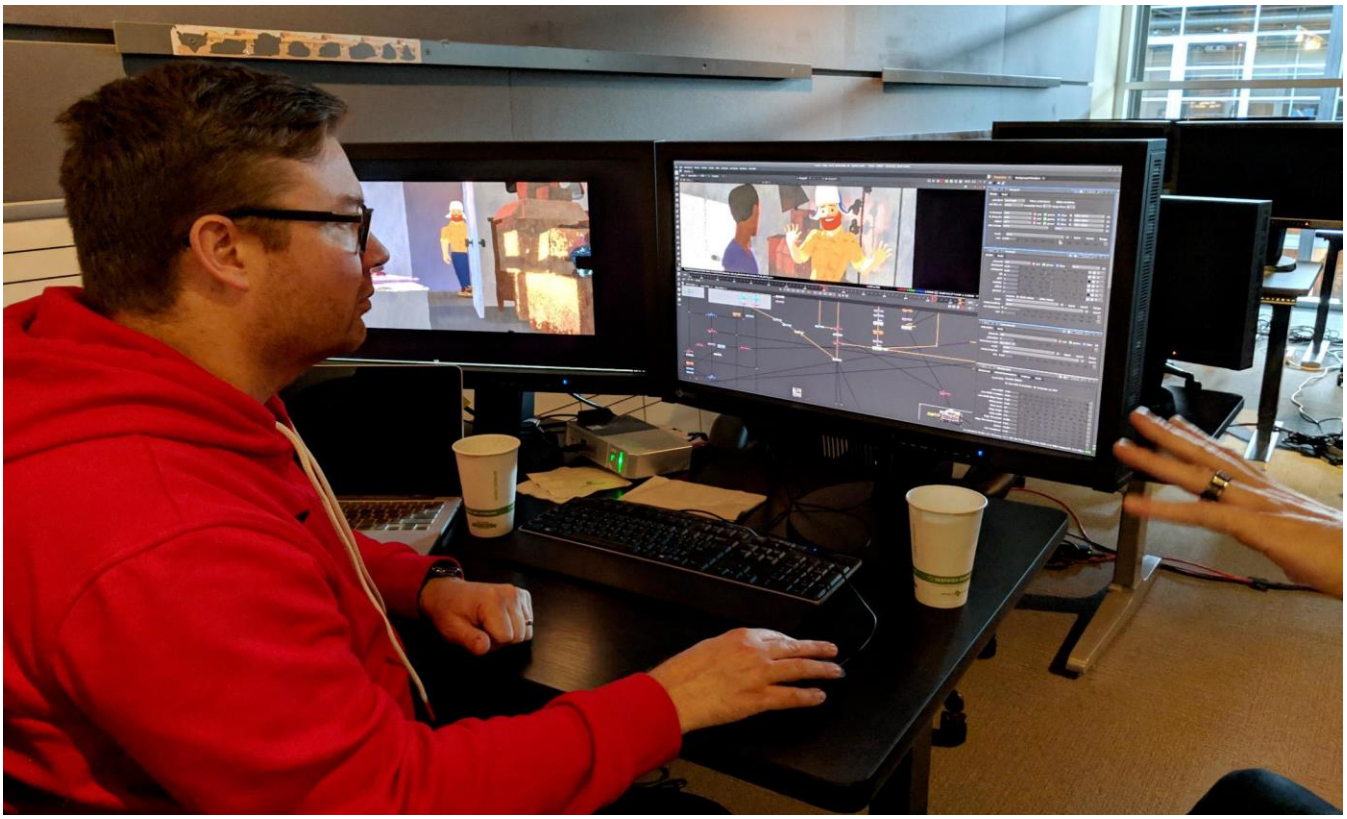



McQueenDemo - cars3 - Presto cars3
PRESTO CREATE EDIT WINDOW HELP LAYOUT
PROJECT
Re-Read modchars_McQueen
Re-Read All Layers
Save
FRAME
SELECT
DMANIP
VISIBILITY
ADD
Movers
Other
Geometry
Weights
Attributes
Transform
Scope
Sphere
Union
Double
Sculpt
ExecMarker
Pivot
PosNeg
String
Name
Value
Set Type
Type
World
ShotPrim
anim
chars
McQueen
Anim
Path
Body
Face
Wheels
Suspension
Kingpin
Collision
Variant
AnimLights
AnimSculpt
Geom
Material
Fix
Kingpin
props
Lattices
sets
main_cam
crowds
sim
fx
mpaint
cameraRigs
prims
important
True
poseName
ReferencePose
makeRefPose
True
makePoseVariant
True
Constraint Axes
Driving System Guides
info
material
other
pipeline
primary
secondary
Warning: Tried to create duplicate command 'CaptureChunk.RevealInShowBrowser' (in CmdRegistry::CreateCommand at menv/Core/lib/Cmd/Registry.cpp:113)
1 Warning
Show Transcript






Pixar's Production Pipeline
Every Pixar movie goes through these steps,
but the process is not entirely linear.
START
Story & Art
Modeling
Rigging
Surfaces
Sets & Cameras
Animation
Simulation
Lighting
Rendering
Complexity can increase render time
Rendered images show work in progress
Joy's surface is also a light source
A model is revised if it looks wrong "on camera"
Texture can be modeled or shaded
Animation needs may require changes to the rig



Overview

Pixar creates animated movies using advanced software.
Although audiences see movies, artists interact daily with complex animation tools.

HCI in Animation Software

Animators require:

- Easy navigation
- Timeline editing
- Drag-and-drop controls
- Real-time preview
- Keyboard shortcuts

Good HCI Improves

- Productivity
- Creativity
- Accuracy
- Collaboration

Example Tasks

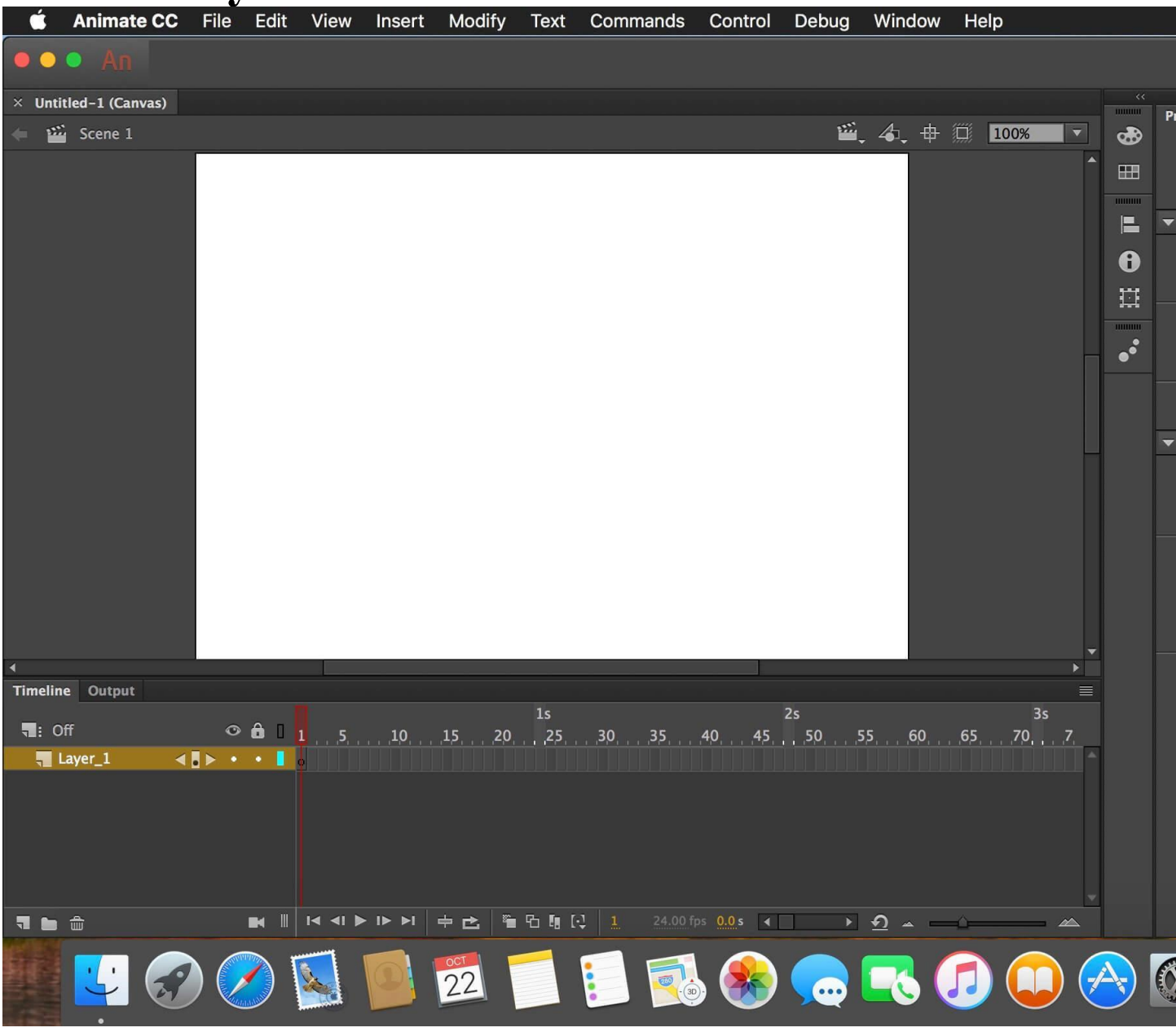
Animator:

1. Select character
2. Move skeleton
3. Preview animation
4. Undo mistakes

5. Save changes

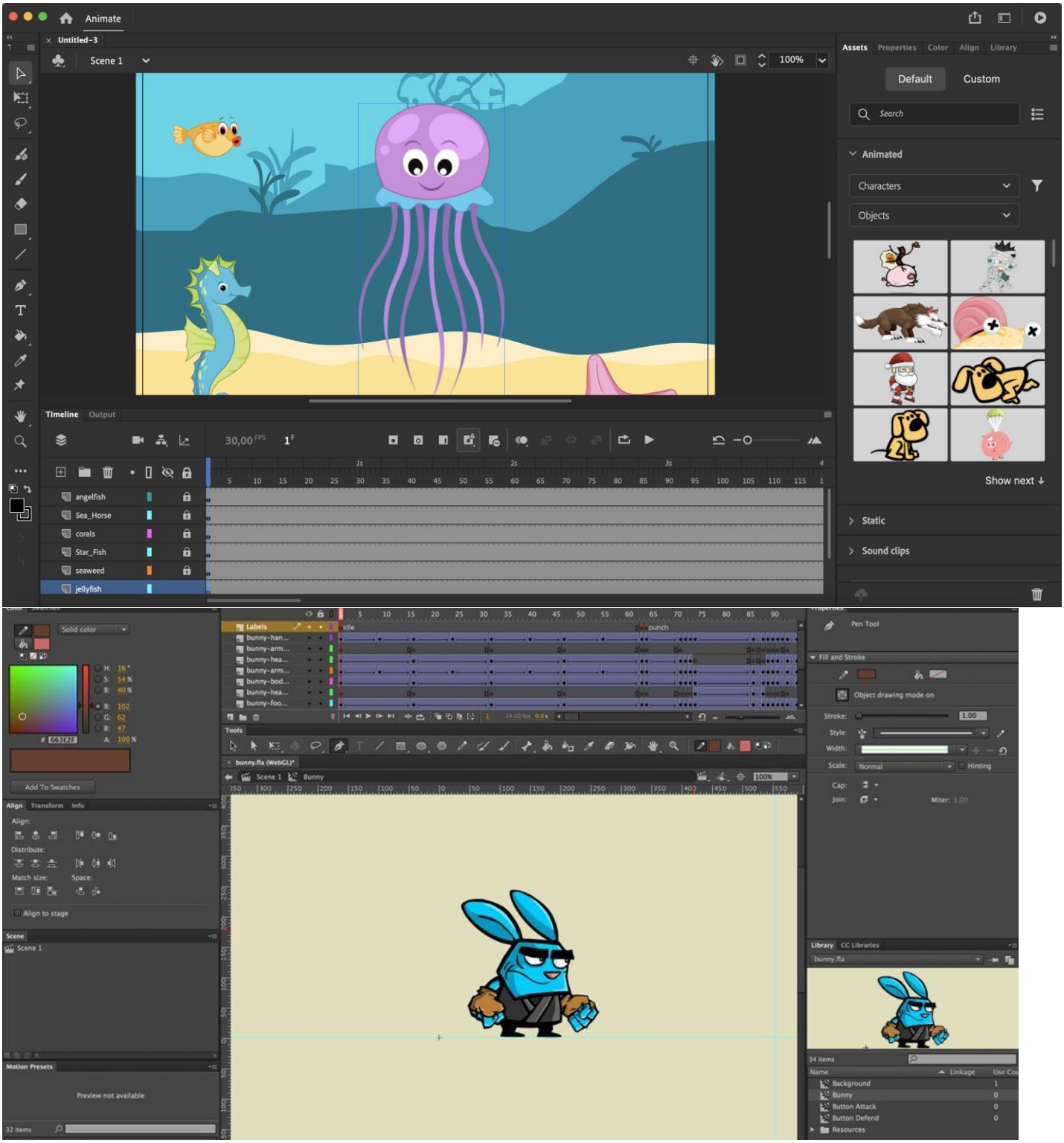
Efficient interfaces reduce workload and errors.

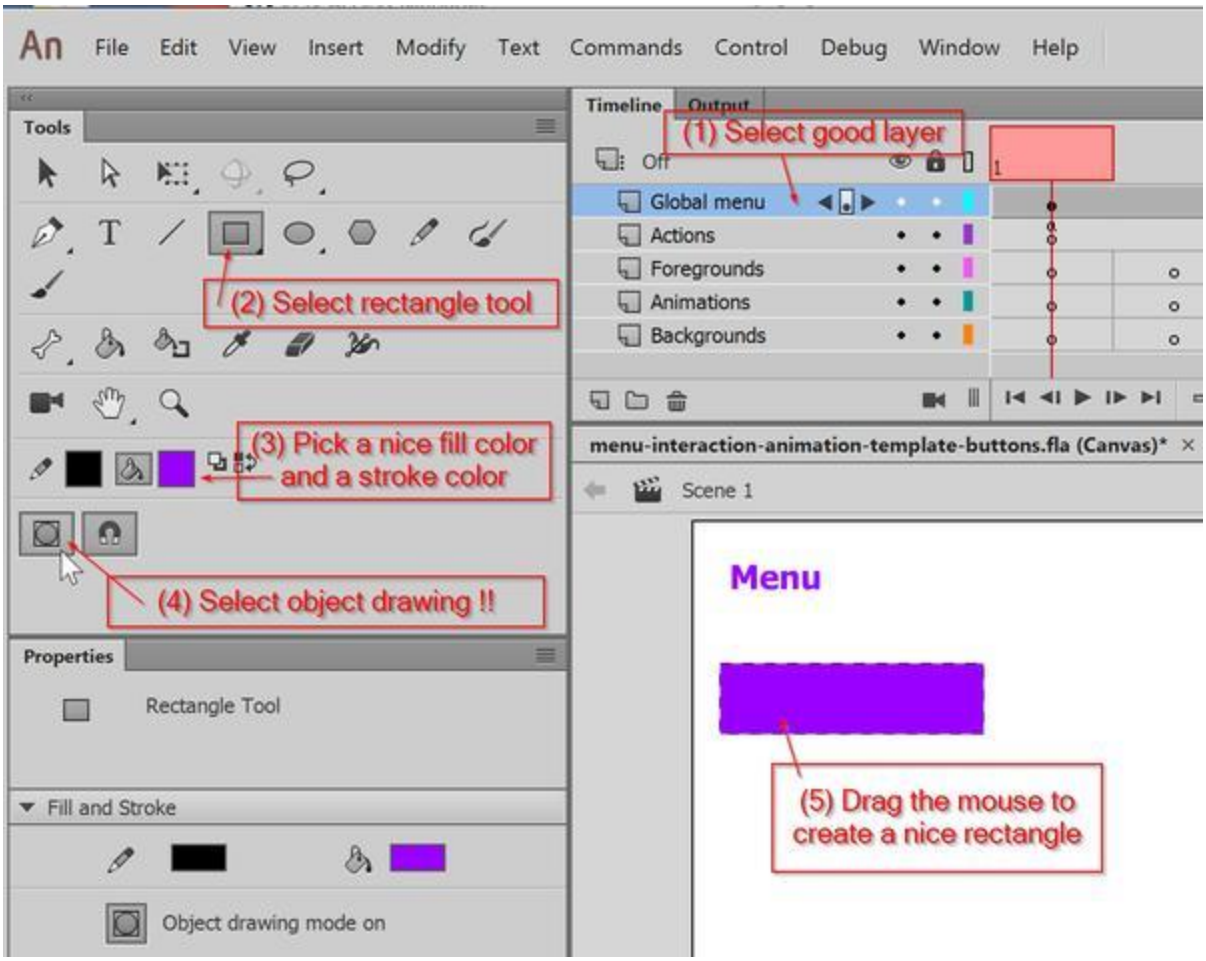
Case Study 6: Adobe Animate




File Edit View Insert Modify Text Commands Control Debug Window Help
Hippo.fla (Canvas)
Scene 1
Timeline
Output
pulse
beam2
beam1
Grass
hippo
ground
moon
sky


Tools


Animate
Untitled-3
Scene 1
100%
Assets
Properties
Color
Align
Library
Default
Custom
Search
Animated
Characters
Objects
Show next
Static
Sound clips
Timeline
Output
30,00 FPS
1 F
angelfish
Sea_Horse
corals
Star_Fish
seaweed
jellyfish
Solid color
Add To Swatches
Align
Transform
Info
Distribute:
Match size:
Space:
Align to stage
Scene
Scene 1
Motion Presets
Preview not available
32 items
Labels
idle
punch
bunny-han...
bunny-arm...
bunny-hea...
bunny-arm...
bunny-bod...
bunny-hea...
bunny-foo...
Tools
bunny.fla (WebGL)*
Scene 1
Bunny
100%
Pen Tool
Fill and Stroke
Object drawing mode on
Stroke:
Style:
Width:
Scale:
Normal
Hinting
Cap:
Join:
Miter:
Library
CC Libraries
bunny.fla
34 items
Name
Linkage
Background
Bunny
Button Attack
Button Defend
Resources



HCI Features

Professional animators benefit from:

- Tool grouping
- Dockable panels
- Keyboard shortcuts
- Layer management
- Timeline visualization
- Undo/Redo
- Drag-and-drop assets

HCI Principles

- Recognition rather than recall
- Flexibility
- User control
- Consistency

Comparison of the Case Studies

Case Study	HCI Focus	Key Principle
Minecraft	Simple UI	Learnability
Fortnite	Competitive UI	Feedback
Pokémon GO	AR Interaction	Natural Mapping
Wii Sports	Motion Controls	Direct Manipulation
Pixar	Professional Tools	Efficiency
Adobe Animate	Animation Software	Flexibility

Common HCI Principles Used in Gaming

Principle	Example
Visibility	Health bar
Feedback	Explosion animation
Consistency	Same controls throughout
Learnability	Tutorial levels
Accessibility	Color-blind mode
User Control	Pause, Undo, Settings
Error Prevention	Confirmation dialogs
Natural Mapping	Motion controls

Challenges in Gaming HCI

- Complex controls
- Information overload
- Small mobile screens
- Motion sickness in VR
- Accessibility for players with disabilities
- Network latency affecting interaction
- Cross-platform consistency

Emerging Trends

Virtual Reality (VR)

- Hand tracking
- Eye tracking
- Immersive interaction

Examples

- Beat Saber
- Half-Life: Alyx

Augmented Reality (AR)
Digital objects are placed in the real world.
Examples:
- Pokémon GO
- Minecraft Earth (discontinued)

Artificial Intelligence (AI)
AI enables:
- Adaptive difficulty
- Smart NPCs
- Personalized recommendations
- Dynamic game environments

Eye Tracking
Tracks where players are looking.
Applications:
- Faster menu selection
- Accessibility
- Foveated rendering in VR

Voice Interaction
Players use voice commands to:
- Control menus
- Issue squad commands
- Search content

Best Practices for HCI in Animation & Gaming
- Keep interfaces simple and uncluttered.
- Provide immediate visual, audio, or haptic feedback.
- Maintain consistent controls and layouts.
- Offer tutorials for beginners.
- Include accessibility features (subtitles, remappable controls, color-blind modes).
- Reduce unnecessary cognitive load.
- Support personalization through adjustable settings.
- Conduct usability testing with representative users before release.

Real-World HCI Lessons

Lesson	Industry Example
Simple interfaces improve usability	Minecraft
Fast feedback enhances engagement	Fortnite
Natural interaction increases immersion	Wii Sports
Context-aware systems enrich experiences	Pokémon GO
Efficient tools improve professional workflows	Pixar
Flexible interfaces support creativity	Adobe Animate
